

December 18, 2025

1. By how much does the emitted current change if the tungsten cathode temperature is increased from $T = 2250 \text{ K}$ to 24000 K ? The work function for tungsten is $W_{\text{ki}} = 5.0 \text{ eV}$. The Richardson coefficient is $6.05 \cdot 10^5 \text{ A/m}^2\text{K}^2$.
2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at $T = 2300 \text{ K}$? The electron charge is $q = -1.6 \cdot 10^{-19} \text{ As}$.
3. What is the free-electron drift velocity in a copper conductor of radius $R = 2.5 \text{ mm}$ if the current is $I = 15 \text{ A}$? The electron density in copper is $n_e = 3.3 \cdot 10^{28} \text{ m}^{-3}$.
4. We examine an n-type semiconductor under the conditions shown in the figure: $B_y = 0.5 \text{ T}$, $U_{\text{H}} = 12 \text{ mV}$, $I = 25 \text{ mA}$. The magnetic induction is directed along the y -axis.
 - (a) What is the electron concentration?
 - (b) What is the drift velocity of the electrons?
 - (c) What is the Hall constant?

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